# PostgreSQL 16 Advanced Indexing & EXPLAIN Plans

**PostgreSQL 16.6**
**VERIFIED RUNTIME**

## KEY SYNTAX & IDIOMS

**Explain Analyze Plan**
```
EXPLAIN (ANALYZE, BUFFERS) SELECT ...;
```
Inspects actual execution runtime costs

**B-Tree Index Creation**
```
CREATE INDEX idx_users_email ON users(email);
```
Accelerates lookups and range scans

**Common Table Expression**
```
WITH ranked AS (SELECT ...) SELECT * FROM ranked;
```
Modularizes complex query steps

**Safe Upsert ON CONFLICT**
```
INSERT INTO ... ON CONFLICT (id) DO UPDATE ...;
```
Atomic insert or update operation

## COMMON PITFALLS & FIXES

**Pitfall:** DROP DATABASE in application code
**Rule:** Strictly block destructive DDL in verification sandboxes

**Pitfall:** SELECT * in high-volume tables
**Rule:** Specify explicit column projections to save buffer reads

**Pitfall:** Sequential scans on foreign keys
**Rule:** Always index foreign key columns used in joins

## TINY VERIFIED EXAMPLE: POSTGRESQL 16 CTE QUERY

**SOURCE CODE:**
```
WITH demo AS (SELECT 'Verified' AS status)
SELECT status FROM demo;
```

**REAL CAPTURED OUTPUT:**
```
status: Verified
```